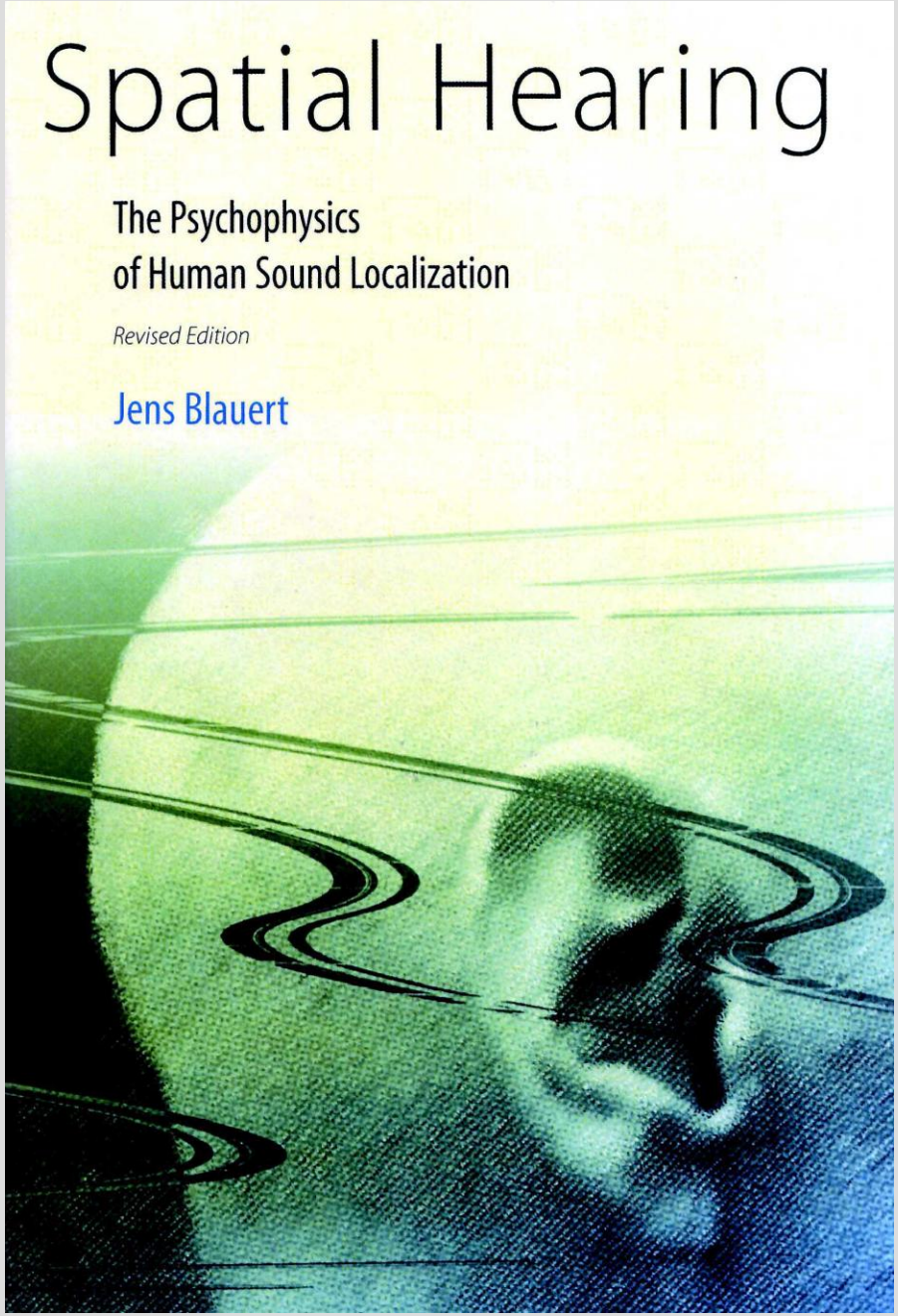


Spatial Hearing

The Psychophysics of
Human Sound Localization

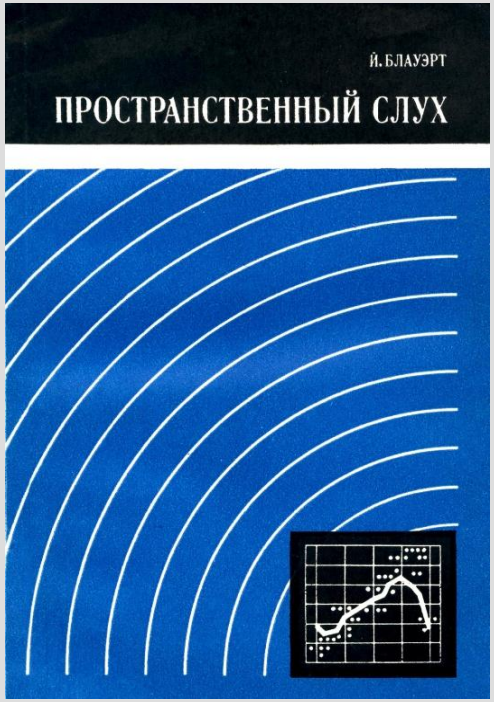
***Jens Blauert* (1997)**

1st edition 1983
2nd, enlarged edition 1997

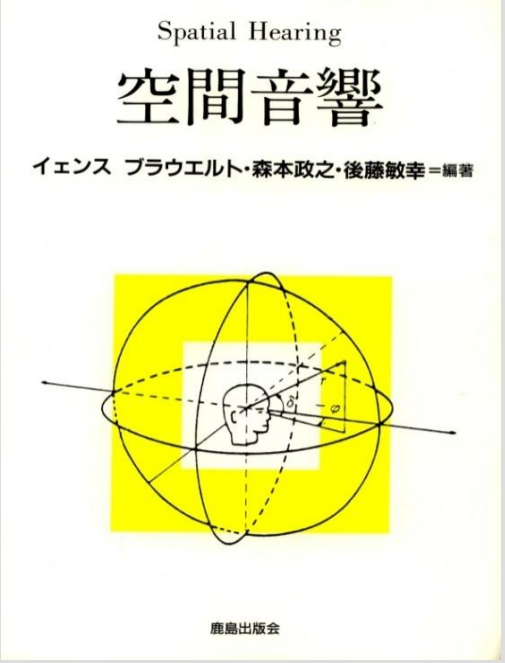
The MIT Press, Harvard MA,
ISBN 0-262-02413-6

(a)

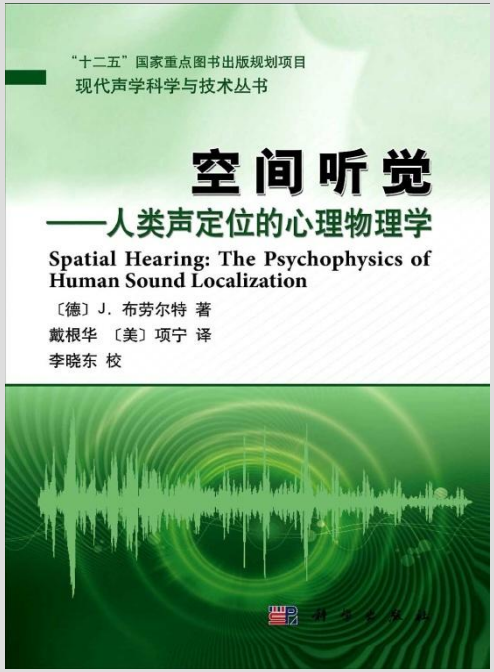
(b)



(c)



(d)



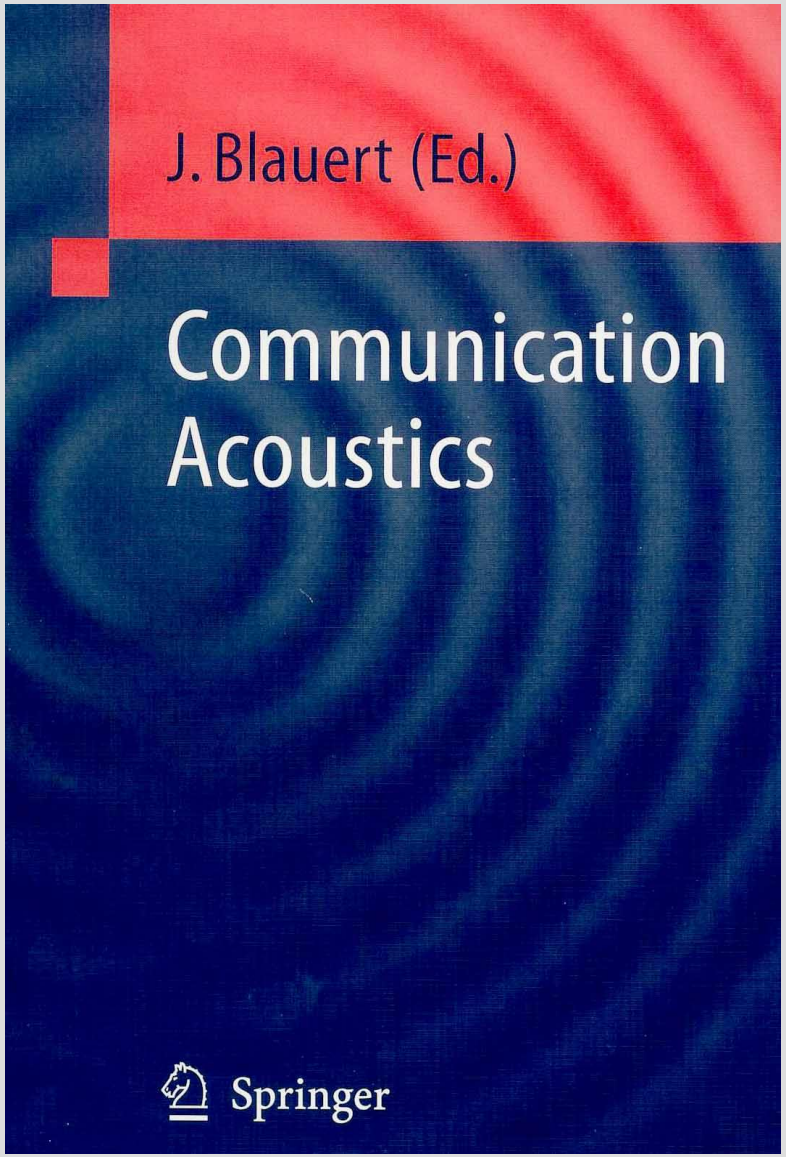
(e)

(a), (b) Spatial Hearing (German) 1974, 2013

(c) Spatial Hearing (Russian) 1979

(d) Spatial Hearing (Japanese) 1986 with M. Morimoto and T. Gotoh

(e) Spatial Hearing (Chinese) 2013



Communication Acoustics

Jens Blauert, ed. (2005)

Authors: *Jens Blauert, Jonas Braasch, Hugo Fastl,Volkmar Hamacher, Dorte Hammershøi, Ulrich Heute, Inga Holube, Herbert Hudde, Ute Jekosch, Georg Klump, Armin Kohlrausch, Arild Lacroix, Henrik Møller, Sebastian Möller, John N. Mourjopoulos, Pedro Novo, Steven van der Par*

Springer Berlin–Heidelberg–New York
ISBN 3-540-22162-X

Communication Acoustics
***Jens Blauert*, ed. (2005)**

Contents

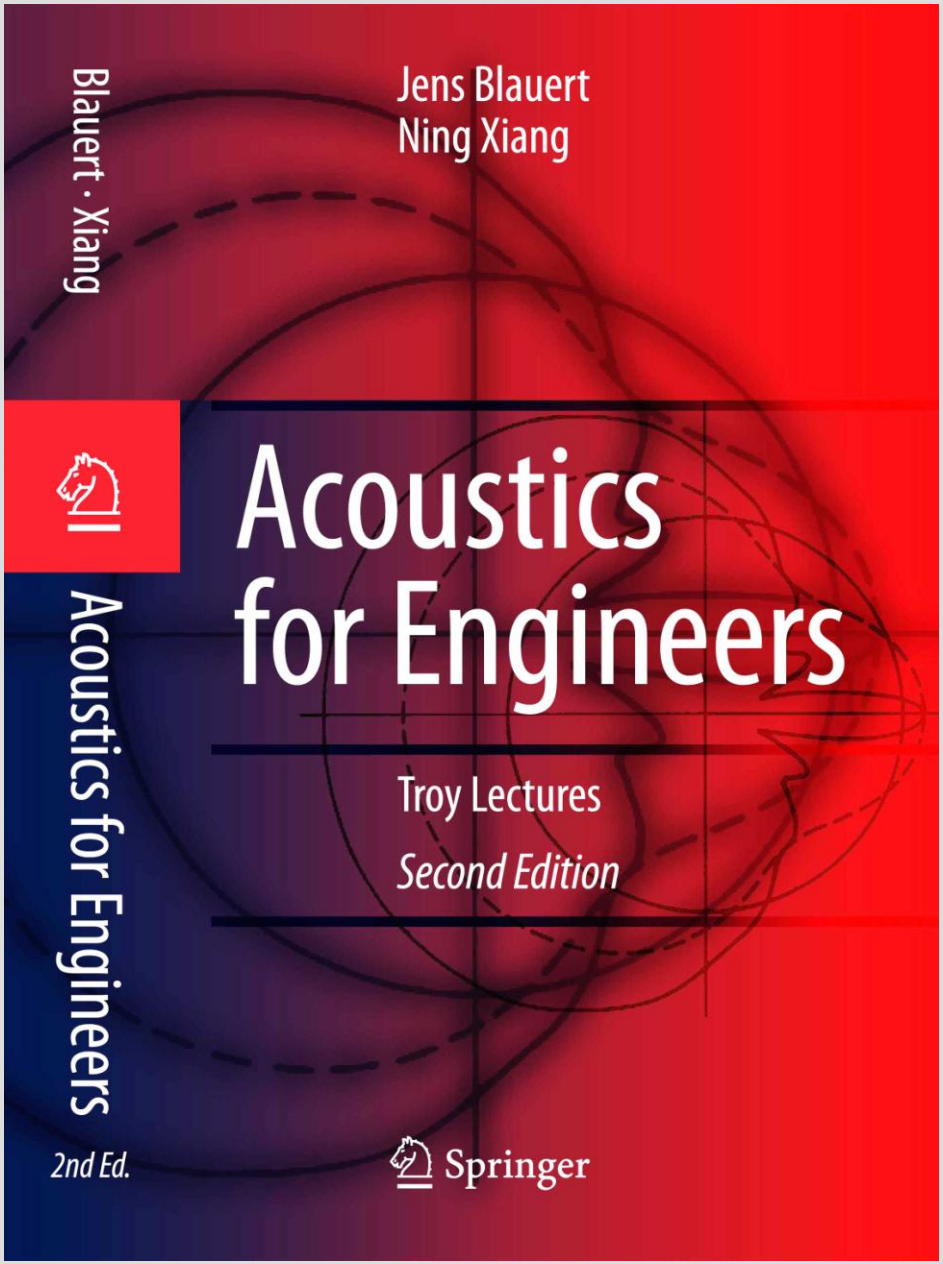
- Blauert, J.: Analysis and synthesis of auditory scenes - Klump, G.: Evolutionary adaptations for auditory communication - Hudde, H.: A functional view on the human hearing Organ - Braasch, J.: Modelling of binaural hearing - Kohlrausch, A. & S. van der Par: Audio-visual interactions in the context of multi-media applications - Fastl, H.: Psycho-acoustics and sound quality - Jekosch, U.: Semiotics for engineers - Möller, S.: Quality of transmitted speech for humans and machines. - Hammershøi, D. & H. Møller: Binaural technique: basic methods for re-cording, synthesis and reproduction - Holube, I. & V. Hamacher: Hearing-aid technology - Novo, P.: Auditory virtual environments - Mourjopoulos, J. N.: The evolution of digital audio technology - Lacroix, A.: Speech-production: acoustics, models and applications - Heute, U. : Speech and audio coding: aiming at high quality and low data rates

(a)

(b)

(a) Communication Acoustics
J. Blauert, ed. (Chinese) 2009

(b) Speech Technology (German)
with E. Schaffert 1985



Acoustics for Engineers

Troy Lectures
– a textbook –

Jens Blauert & Ning Xiang

1st edition 2008
2nd, enlarged edition 2009

Springer Berlin–Heidelberg–New York
ISBN 978-3-540-76346-8

Jens Blauert & Ning Xiang

Acoustics for Engineers –Troy Lectures (2nd ed.)

Springer Berlin–Heidelberg–New York (2009)

Contents

Introduction

1. **Mechanic and acoustic oscillators**
2. **Electromechanic and electroacoustic oscillations**
3. **Electromechanic and electroacoustic analogies**
4. **Electromechanic and electroacoustic transduction**
5. **Magnetic-field transducers**
6. **Electric-field transducers**
7. **The wave equation in fluids**
8. **Horns and stepped ducts**
9. **Spherical sound sources and line arrays**
10. **Piston membranes, diffraction and scattering**
11. **Dissipation, reflection and absorption**
12. **Geometric acoustics and diffuse sound fields**
13. **Isolation of air- and structure-borne sound**
14. **Noise control: a survey**

Appendices, incl. complex notation for sinusoidal signals, and for power and intensity, and a selection of problems

Modern Acoustics and Signal Processing

Jens Blauert *Editor*

The Technology of Binaural Listening

This book reports on the application of advanced models of the human binaural hearing system in modern technology, among others, in the following areas: binaural analysis of aural scenes, binaural de-reverberation, binaural quality assessment of audio channels, loudspeakers and performance spaces, binaural perceptual coding, binaural processing in hearing aids and cochlea implants, binaural systems in robots, binaural/tactile human-machine interfaces, speech-intelligibility prediction in rooms and/or multi-speaker scenarios. An introduction to binaural modeling and an outlook to the future are provided. Further, the book features a MATLAB toolbox to enable readers to construct their own dedicated binaural models on demand.

2013

Engineering

ISBN 978-3-642-37761-7

9 783642 377617

springer.com

Blauert *Ed.*

The Technology of Binaural Listening

Modern Acoustics and Signal Processing

Jens Blauert *Editor*

The Technology of Binaural Listening

ASA Press

Modern Acoustics and Signal Processing

Springer

THE TECHNOLOGY OF BINAURAL LISTENING

ISBN 978-3-642-37761-7

Editor: Jens Blauert – Publisher: Springer/ASA – Publication Year: 2013

Table of Contents

Chap1 **An introduction to binaural processing**
A. Kohlrausch, J. Braasch, D. Kolossa and J. Blauert

Chap2 **The auditory modeling toolbox**
P. Søndergaard and P. Majdak.

Chap3 **Trends in acquisition of individual head-related transfer functions**
G. Enzner, C. Antweiler and S. Spors

Chap4 **Assessment of sagittal-plane sound-localization performance in spatial-audio applications**
R. Baumgartner, P. Majdak and B. Laback

Chap5 **Modeling horizontal localization of complex sounds in the impaired and aided impaired auditory system**
N. Le Goff, J. Buchholz and T. Dau

Chap6 **Binaural scene analysis with multi-dimensional statistical filters**
C. Spille, B. Meyer, M. Dietz and V. Hohmann

Chap7 **Extracting sound-source-distance information from binaural signals**
E. Georganti, T. May, S. van de Par and J. Mourjopoulos

Chap8 **A binaural model that analyses acoustic spaces and stereophonic reproduction systems by utilizing head rotations**
J. Braasch, S. Clapp, A. Pars, T. Pastore and N. Xiang

Chap9 **Binaural systems in robotics**
S. Argentieri, A. Portello, M. Bernard, P. Danès, and B. Gas

Chap10 **Binaural assessment of multi-channel reproduction**
H. Wierstorf, A. Raake and S. Spors

Chap11 **Optimization of binaural algorithms for maximum predicted speech intelligibility**
A. Schlesinger and C. Luther

Chap12 **Modeling sound localization with cochlear implants**
M. Nicoletti, C. Wirtz and W. Hemmert

Chap13 **Binaural assessment of parametrically coded spatial audio signals**
M. Takanen, O. Santala and V. Pulkki

Chap14 **Binaural dereverberation**
A. Tsilfidis, A. Westerman, J. Buchholz, E. Georganti, and J. Mourjopoulos

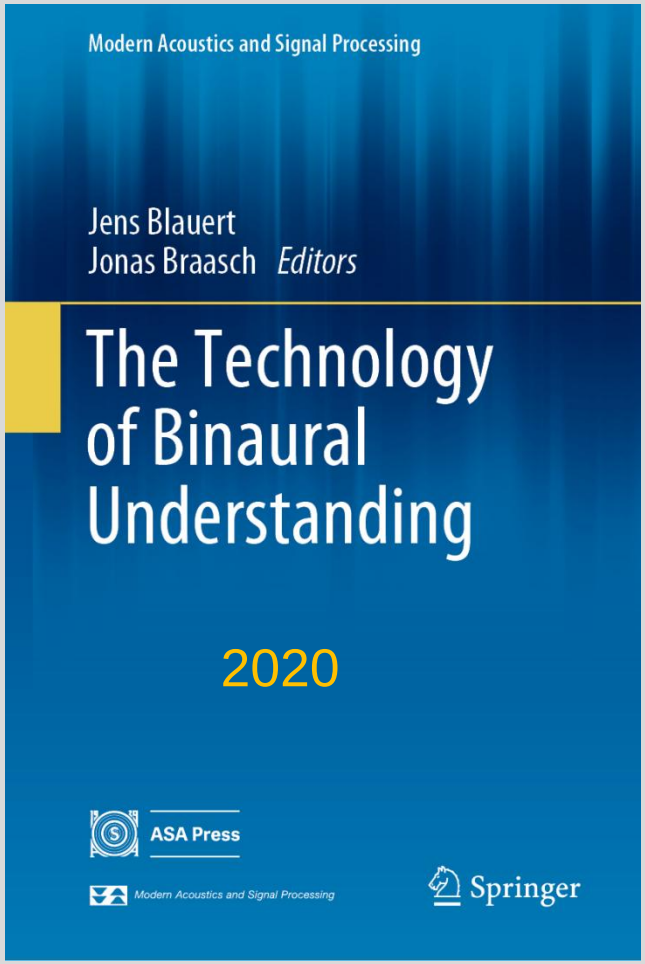
Chap15 **Binaural localization and detection of speakers in complex acoustic scenes**
T. May, S. van de Par and A. Kohlrausch

Chap16 **Predicting binaural speech intelligibility in architectural acoustics**
J. Culling, M. Lavandier and S. Jelfs

Chap17 **Employing binaural–proprioceptive interaction in human-machine interfaces**
M. Stamm and M. Altinsoy

Chap18 **Further challenges and the road ahead**
J. Blauert, D. Kolossa, K. Obermayer, and K. Adiloglu

Published in 2013 as the first book of ASA Press



Part 1: Forming and Interpreting Aural Objects: Effects and Models

Reflexive and Reflective Auditory Feedback
Jens Blauert and Guy J. Brown

Auditory Gestalt Rules and their Application
Sarinah Sutojo, Joachim Thiemann, Armin Kohlrausch and Steven van de Par

Selective Binaural Attention and Attention Switching
Janina Fels, Josefa Oberem and Iring Koch

Blackboard Systems for Cognitive Audition
Christopher Schymura and Dorothea Kolossa

Part 2: Configuring and Understanding Aural-Space

Formation of Three-Dimensional Auditory Space
Piotr Majdak, Robert Baumgartner and Claudia Jenny

Biological Aspects of Perceptual Space Formation
Michael Pecka, Christian Leibold and Benedikt Grothe

Auditory Spatial Impression in Concert Halls
Tapio Lokki and Jukka Pätynen

Auditory Room Learning and Adaptation to Sound Reflections
Bernhard U. Seeber and Samuel Clapp

Room Effect on Musicians' Performance
Malte Kob, Sebastià V. Amengual Garí and Zora Schärer Kalkandjiev

Binaural Modeling from an Evolving Habitat Perspective
Jonas Braasch

Part 3: Processing Cross-Modal Inference

Psychophysical Models of Sound Localisation with Audiovisual
Catarina Mendonça

Cross-Modal and Cognitive Processes in Sound Localization
M. Torben Pastore, Yi Zhou and William A. Yost

Spatial Soundscape Superposition and Multimodal Interaction
Michael Cohen and William L. Martens

Part 4 Evaluating Aural-Scene Quality and Speech Understanding

Binaural Evaluation of Sound Quality and Quality of Experience
Alexander Raake and Hagen Wierstorf

The Language of Rooms: From Perception to Cognition to Aesthetic Judgment
Stefan Weinzierl, Steffen Lepa and Martin Thiering

Modeling the Aesthetics of Audio Scene Reproduction
John Mourjopoulos

A Virtual Testbed for Binaural Agents
Jens Blauert

Binaural Technology for Machine Speech Recognition and Understanding
Richard M. Stern and Anjali Menon

Modeling Binaural Speech Understanding in Complex Situations
Mathieu Lavandier and Virginia Best

Part 5: Applying Cognitive Mechanisms to Audio Technology

Creating Auditory Illusions with Spatial-Audio Technologies
Rozenn Nicol

Creating Auditory Illusions with Binaural Technology
Karlheinz Brandenburg, Florian Klein, Annika Neidhardt,Ulrike Sloma and Stephan Werner

Toward Cognitive Usage of Binaural Displays
Yoiti Suzuki, Akio Honda, Yukio Iwaya, Makoto Ohuchi and Shuichi Sakamoto

Audition as a Trigger of Head Movements
Benjamin Cohen-Lhyver, Sylvain Argentieri and Bruno Gas

Intelligent Hearing Instruments—Trends and Challenges
Eleftheria Georganti, Gilles Courtois, Peter Derleth and Stefan Launer

Scene-Aware Dynamic Range Compression in Hearing Aids
Tobias May, Borys Kowalewski and Torsten Dau